

Hospital Management System



Project Documentation

Submitted To

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Introduction

The Hospital Management System is a database project developed to efficiently manage hospital-related information using a relational database. The system is designed to store and organize records of patients, doctors, departments, appointments, admissions, medicines, prescriptions, billing, staff, and medical records in a structured manner.

The main purpose of this project is to replace manual record-keeping with a computerized database that ensures data accuracy, consistency, and easy access to information. By using SQL, different tables are connected through primary keys and foreign keys to maintain relationships between hospital entities and reduce data redundancy.

This project demonstrates the practical implementation of database concepts such as table creation, constraints, relationships, normalization, and SQL queries. It provides an organized way to manage hospital data while allowing users to insert, update, delete, and retrieve information efficiently.

Overall, the Hospital Management System improves data management, supports faster access to records, and ensures the integrity of hospital information through a well-designed relational database.

Project Objectives

The main objectives of the Hospital Management System are:

- To develop a relational database for managing hospital operations efficiently.
- To maintain accurate records of patients, doctors, departments, appointments, admissions, staff, medicines, prescriptions, billing, and medical records.
- To reduce data redundancy by organizing information into related tables.
- To establish relationships between tables using primary keys and foreign keys.
- To ensure data integrity through appropriate database constraints.
- To provide an efficient system for storing, updating, deleting, and retrieving hospital data using SQL queries.

- To demonstrate the practical application of database management concepts, including table creation, relationships, normalization, and data manipulation.
- To improve the organization and accessibility of hospital information through a structured database system.

Project Scope

The Hospital Management System is designed to manage and organize essential hospital data within a relational database. The project focuses on storing, maintaining, and retrieving information related to various hospital operations through SQL.

The database includes records of patients, doctors, departments, appointments, admissions, staff members, medicines, prescriptions, billing, and medical records. These entities are connected through primary keys and foreign keys to ensure data consistency and maintain proper relationships between tables.

The system allows users to:

- Store and manage hospital-related information in a structured database.
- Insert, update, delete, and retrieve records using SQL queries.
- Maintain relationships between different entities using foreign keys.
- Ensure data accuracy and integrity through constraints.
- Generate meaningful information by querying the database.

This project is limited to the database implementation only. It does not include a graphical user interface (GUI) or web application. Its primary purpose is to demonstrate the design and implementation of a relational database for hospital management using SQL.

System Modules

The Hospital Management System is divided into different modules, where each module is responsible for managing a specific part of the hospital database.

1. Departments Module
2. Doctors Module
3. Patients Module
4. Appointments Module
5. Admissions Module
6. Pharmacy Module
7. Prescription Module
8. Billing Module
9. Staff Module
10. Laboratory Module
11. Rooms Module

Database Design

The Hospital Management System is implemented using a relational database model. The database is designed to store hospital information in separate but interconnected tables, ensuring data consistency, minimizing redundancy, and maintaining data integrity. Relationships between tables are established using primary keys and foreign keys.

The database consists of the following tables:

Table Name	Description
Department	Stores information about hospital departments.
Doctor	Stores details of doctors, including their department.
Patient	Maintains patient personal and contact information.
Appointment	Records appointments between patients and doctors.
Admission	Stores patient admission details.
Pharmacy	Maintains information about medicines available in the hospital.
Prescription	Stores medicine prescriptions issued to patients by doctors.

Billing	Maintains billing and payment information for patients.
Staff	Stores details of hospital staff members.
Laboratory	Stores laboratory test details, including test names, results, and test dates for patients.
Rooms	Maintains information about hospital rooms, including room type and availability status.

Each table has a primary key that uniquely identifies every record. Foreign keys are used to establish relationships between related tables, ensuring referential integrity and preventing invalid data entries. The database structure follows relational database principles, making data storage efficient and simplifying data retrieval through SQL queries.

Departments Table

Description

The Departments table stores information about the different departments in the hospital, such as Cardiology, Neurology, and Orthopedics. It acts as the parent table for doctors, allowing each doctor to be associated with a specific department.

Constraints

- PRIMARY KEY (Department_ID)
- AUTO_INCREMENT (Department_ID)
- NOT NULL (Department_Name)

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:				
	Department_ID	Department_Name	Floor	Description
▶	1	Cardiology	First Floor	Heart Department
	2	Neurology	Second Floor	Brain Department
	3	Orthopedics	Third Floor	Bone Department
	4	Pediatrics	Second Floor	Children Department
	5	Emergency	Ground Floor	Emergency Cases
	6	Dermatology	Second Floor	Skin Care Department
	7	ENT	First Floor	Ear, Nose and Throat Department
	8	Gynecology	Fourth Floor	Women Health and Maternity Department
	9	Oncology	Third Floor	Cancer Treatment Department
	10	Radiology	Basement	X-Ray, MRI and CT Scan Department
*	NULL	NULL	NULL	NULL

Doctors Table

Description

The Doctors table stores information about doctors working in the hospital, including their personal details, qualifications, specialization, and contact information. Each doctor is assigned to a specific department through a foreign key, allowing efficient management of departmental records.

Constraints

- PRIMARY KEY (Doctor_ID)
- AUTO_INCREMENT (Doctor_ID)
- FOREIGN KEY (Department_ID)
- NOT NULL (Doctor_Name, Gender, Qualification, Specialization, Phone, Email, Department_ID)
- UNIQUE (Phone)
- CHECK (Gender IN ('Male', 'Female'))

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:							
	Doctor_ID	Doctor_Name	Gender	Qualification	Specialization	Phone	Department_ID
▶	101	Ahmed Khan	Male	MBBS	Cardiologist	03001234567	1
	102	Sara Ali	Female	MBBS	Neurologist	03111234567	2
	103	Usman Tariq	Male	MBBS	Orthopedic Surgeon	03221234567	3
	104	Ayesha Noor	Female	MBBS	Pediatrician	03331234567	4
	105	Ali Raza	Male	MBBS	General Physician	03441234567	5
	106	Hina Ahmed	Female	MBBS	Dermatologist	03061112222	6
	107	Umar Farooq	Male	MBBS	ENT Specialist	03072223333	7
	108	Ayesha Noor	Female	MBBS	Gynecologist	03083334444	8
	109	Hassan Raza	Male	MBBS	Oncologist	03094445555	9
	110	Sana Iqbal	Female	MBBS	Radiologist	03105556666	10
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Patients Table

Description

The Patients table maintains complete information about patients, including their personal details, blood group, disease, contact number, and address. It serves as the main table that is linked with appointments, admissions, prescriptions, billing, and laboratory records.

Constraints

- PRIMARY KEY (Patient_ID)
- AUTO_INCREMENT (Patient_ID)
- NOT NULL (Patient_Name, Gender, Age, Blood_Group, Phone, Address, Disease)
- UNIQUE (Phone)
- CHECK (Age > 0)
- CHECK (Gender IN ('Male', 'Female'))

Result Grid			 Filter Rows:		Edit:				Export/Import:			Wrap Cell Content:	
	Patient_ID	Patient_Name	Gender	Age	Blood_Group	Phone	Address	Disease					
▶	201	Hassan Ali	Male	22	A+	03001234567	Lahore	Fever					
	202	Fatima Noor	Female	25	B+	03111234567	Karachi	Migraine					
	203	Bilal Ahmed	Male	30	O+	03221234567	Islamabad	Fracture					
	204	Ayesha Khan	Female	28	AB+	03331234567	Multan	Diabetes					
	205	Hamza Malik	Male	35	A-	03441234567	Faisalabad	Asthma					
	206	Areeba Khan	Female	22	B+	03011112222	Lahore	Skin Allergy					
	207	Hamza Ali	Male	31	O+	03022223333	Islamabad	Sinus Infection					
	208	Zainab Fatima	Female	27	A-	03033334444	Faisalabad	Pregnancy Checkup					
	209	Bilal Hussain	Male	45	AB+	03044445555	Multan	Lung Cancer					
	210	Noor Ul Ain	Female	29	O-	03055556666	Rawalpindi	Back Pain					
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL					

Appointments Table

Description

The Appointments table stores appointment details between patients and doctors. It records the appointment date, time, and current appointment status, enabling efficient scheduling and management.

Constraints

- PRIMARY KEY (Appointment_ID)
- AUTO_INCREMENT (Appointment_ID)
- FOREIGN KEY (Patient_ID, Doctor_ID)
- NOT NULL (Patient_ID, Doctor_ID, Appointment_Date, Appointment_Time, Status)
- DEFAULT ('Scheduled') for Status

Appointment_ID	Patient_ID	Doctor_ID	Appointment_Date	Appointment_Time	Status
401	201	101	2026-07-20	10:00:00	Completed
402	202	102	2026-07-21	11:30:00	Pending
403	203	103	2026-07-22	09:00:00	Completed
404	204	104	2026-07-23	02:00:00	Cancelled
405	205	105	2026-07-24	12:30:00	Pending
411	206	106	2026-07-10	09:00:00	Scheduled
412	207	107	2026-07-10	10:00:00	Completed
413	208	108	2026-07-11	11:00:00	Scheduled
414	209	109	2026-07-11	02:00:00	Cancelled
415	210	110	2026-07-12	03:30:00	Scheduled
NULL	NULL	NULL	NULL	NULL	NULL

Admissions Table

Description

The Admissions table keeps records of patients admitted to the hospital. It stores admission and discharge dates while linking patients with their assigned rooms.

Constraints

- PRIMARY KEY (Admission_ID)
- AUTO_INCREMENT (Admission_ID)
- FOREIGN KEY (Patient_ID, Room_No)
- NOT NULL (Patient_ID, Room_No, Admit_Date, Discharge_Date)

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
	Admission_ID	Patient_ID	Room_No	Admit_Date	Discharge_Date
▶	301	201	101	2026-07-18	2026-07-22
	302	202	102	2026-07-19	2026-07-24
	303	203	103	2026-07-20	2026-07-25
	304	204	104	2026-07-21	2026-07-23
	305	205	105	2026-07-22	2026-07-26
	306	206	106	2026-07-10	2026-07-13
	307	207	105	2026-07-11	2026-07-15
	308	208	108	2026-07-12	2026-07-16
	309	209	101	2026-07-13	2026-07-18
	310	210	110	2026-07-14	2026-07-17
*	NULL	NULL	NULL	NULL	NULL

Pharmacy Table

Description

The Pharmacy table stores information about medicines available in the hospital pharmacy, including medicine name, price, and stock quantity. It helps in maintaining medicine inventory.

Constraints

- PRIMARY KEY (Medicine_ID)
- AUTO_INCREMENT (Medicine_ID)
- NOT NULL (Medicine_Name, Price, Stock)
- CHECK (Price >= 0)
- CHECK (Stock >= 0)

Result Grid						
Filter Rows:						
Edit:						
Export/Import:						
	Medicine_ID	Medicine_Name	Company	Quantity	Price	Expiry_Date
▶	501	Panadol	GSK	100	120.00	2027-05-15
	502	Augmentin	GSK	50	450.00	2027-08-20
	503	Brufen	Abbott	80	180.00	2028-01-10
	504	Disprin	Bayer	120	90.00	2027-11-30
	505	Flagyl	Sanofi	60	250.00	2028-03-18
	506	Paracetamol	GSK Pakistan	200	50.00	2028-06-30
	507	Amoxicillin	Getz Pharma	120	180.00	2027-12-31
	508	Ibuprofen	Highnoon Laboratories	150	80.00	2028-03-31
	509	Omeprazole	Sami Pharmaceuticals	90	250.00	2027-10-31
	510	Cetirizine	Hilton Pharma	300	40.00	2028-08-31
*	NULL	NULL	NULL	NULL	NULL	NULL

Prescriptions Table

Description

The Prescriptions table stores medicines prescribed by doctors to patients. It records the prescribed medicine, dosage, and duration of treatment while maintaining relationships between doctors and patients.

Constraints

- PRIMARY KEY (Prescription_ID)
- AUTO_INCREMENT (Prescription_ID)
- FOREIGN KEY (Patient_ID, Doctor_ID)
- NOT NULL (Patient_ID, Doctor_ID, Medicine_Name, Dosage, Duration)

Prescription_ID	Patient_ID	Doctor_ID	Medicine_Name	Dosage	Duration
601	201	101	Panadol	1 Tablet Twice Daily	5 Days
602	202	102	Augmentin	1 Capsule Daily	7 Days
603	203	103	Brufen	1 Tablet After Meal	3 Days
604	204	104	Disprin	1 Tablet Daily	5 Days
605	205	105	Flagyl	2 Tablets Daily	7 7 Days
606	206	106	Cetirizine	1 Tablet Twice Daily	7 Days
607	207	107	Amoxicillin	1 Capsule Three Times Daily	5 Days
608	208	108	Paracetamol	1 Tablet Twice Daily	3 Days
609	209	109	Ibuprofen	1 Tablet After Meal	10 Days
610	210	110	Omeprazole	1 Capsule Before Breakfast	14 Days
NULL	NULL	NULL	NULL	NULL	NULL

Billing Table

Description

The Billing table manages patient billing information, including total amount, payment method, and payment status. It ensures accurate billing records for hospital services.

Constraints

- PRIMARY KEY (Bill_ID)
- AUTO_INCREMENT (Bill_ID)
- FOREIGN KEY (Patient_ID)

- NOT NULL (Patient_ID, Total_Amount, Payment_Method, Payment_Status)
- DEFAULT ('Pending') for Payment_Status
- CHECK (Total_Amount >= 0)

Result Grid						
Filter Rows:						
	Bill_ID	Patient_ID	Total_Amount	Payment_Method	Payment_Status	Bill_Date
▶	801	201	2500.00	Cash	Paid	2026-07-01
	802	202	4000.00	Card	Pending	2026-07-02
	803	203	3500.00	Cash	Paid	2026-07-03
	804	204	5000.00	Online	Pending	2026-07-04
	805	205	2800.00	Cash	Paid	2026-07-05
	806	206	2500.00	Cash	Paid	2026-07-10
	807	207	4500.00	Card	Paid	2026-07-10
	808	208	3200.00	Online	Pending	2026-07-11
	809	209	15000.00	Card	Paid	2026-07-11
	810	210	2800.00	Cash	Pending	2026-07-12
*	NULL	NULL	NULL	NULL	NULL	NULL

Laboratory Table

Description

The Laboratory table stores laboratory test information for patients, including test name, test results, and test date. It helps maintain diagnostic records for future reference.

Constraints

- PRIMARY KEY (Test_ID)
- AUTO_INCREMENT (Test_ID)
- FOREIGN KEY (Patient_ID)
- NOT NULL (Patient_ID, Test_Name, Test_Result, Test_Date)

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
	Test_ID	Patient_ID	Test_Name	Test_Date	Result
▶	701	201	Blood Test	2026-07-20	Normal
	702	202	MRI Scan	2026-07-21	Migraine Detected
	703	203	X-Ray	2026-07-22	Fracture Confirmed
	704	204	Sugar Test	2026-07-23	High Sugar Level
	705	205	Chest X-Ray	2026-07-24	Normal
	706	206	Skin Allergy Test	2026-07-10	Positive
	707	207	CT Scan	2026-07-10	Normal
	708	204	Ultrasound	2026-07-11	Normal Pregnancy
	709	209	Biopsy	2026-07-11	Cancer Cells Detected
	710	201	X-Ray	2026-07-12	Minor Disc Problem
*	NULL	NULL	NULL	NULL	NULL

Rooms Table

Description

The Rooms table contains information about hospital rooms, including room type and availability status. It supports patient admission by tracking available and occupied rooms.

Constraints

- PRIMARY KEY (Room_No)
- AUTO_INCREMENT (Room_No)
- NOT NULL (Room_Type, Availability)
- DEFAULT ('Available') for Availability

Result Grid				
Filter Rows:				
Edit:				
	Room_No	Room_Type	Charges	Availability
▶	101	General	2000.00	Available
	102	Private	5000.00	Occupied
	103	ICU	10000.00	Available
	104	General	2000.00	Occupied
	105	Private	5000.00	Available
	106	General Ward	1500.00	Available
	107	Private	4000.00	Occupied
	108	ICU	8000.00	Available
	109	Semi-Private	3000.00	Occupied
	110	VIP	10000.00	Available
*	NULL	NULL	NULL	NULL

Staff Table

Description

The Staff table stores information about hospital staff members, including their personal details, position, salary, and contact number. It helps manage hospital employees efficiently.

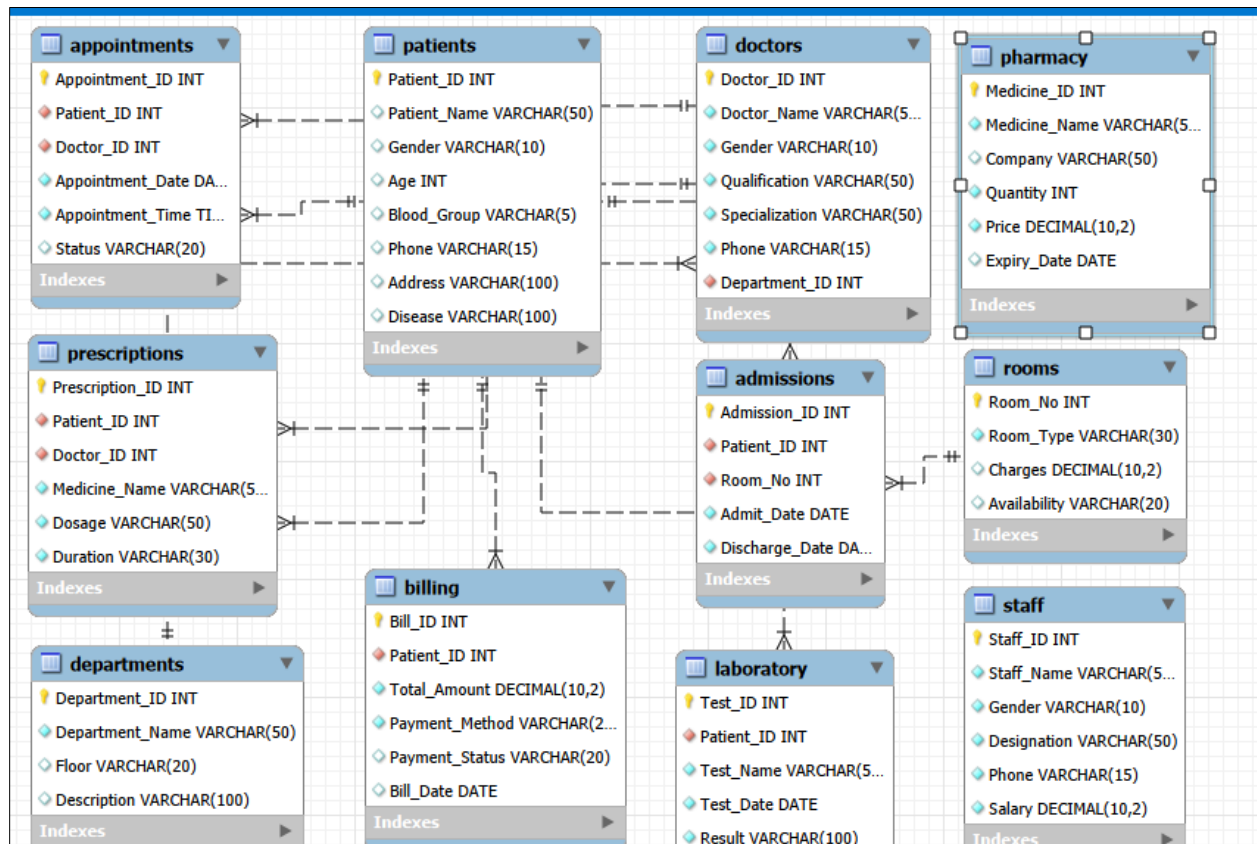
Constraints

- PRIMARY KEY (Staff_ID)
- AUTO_INCREMENT (Staff_ID)
- NOT NULL (Staff_Name, Gender, Phone, Position, Salary)
- UNIQUE (Phone)
- CHECK (Salary >= 0)

Result Grid			 Filter Rows:	Edit: 		  Export/Import:
	Staff_ID	Staff_Name	Gender	Designation	Phone	Salary
▶	901	Maria Khan	Female	Nurse	03001112222	40000.00
	902	Ahmed Raza	Male	Receptionist	03112223333	35000.00
	903	Sana Ali	Female	Lab Technician	03223334444	45000.00
	904	Usman Tariq	Male	Pharmacist	03334445555	50000.00
	905	Hina Noor	Female	Ward Assistant	03445556666	30000.00
	906	Ali Raza	Male	Receptionist	03011110001	45000.00
	907	Fatima Noor	Female	Nurse	03011110002	60000.00
	908	Ahmed Khan	Male	Lab Technician	03011110003	55000.00
	909	Ayesha Malik	Female	Pharmacist	03011110004	70000.00
	910	Usman Tariq	Male	Accountant	03011110005	65000.00
✱	NULL	NULL	NULL	NULL	NULL	NULL

ER Diagram

The Entity Relationship (ER) Diagram represents the structure of the Hospital Management System database. It illustrates the entities, their attributes, and the relationships established through primary and foreign keys. The diagram helps visualize how different tables are connected to ensure data consistency and integrity.



Conclusion

The Hospital Management System database was successfully designed and implemented using SQL. The project demonstrates the practical use of relational database concepts, including table creation, relationships, constraints, and data manipulation. By organizing hospital data into interconnected tables, the system improves data consistency, minimizes redundancy, and enables efficient management of hospital records.

